

SERIAL INTERFACE

The AD5426/AD5432/AD5443 have an easy to use 3-wire interface that is compatible with SPI/QSPI/MICROWIRE and DSP interface standards. Data is written to the device in 16 bit words. This 16-bit word consists of 4 control bits and either 8, 10, or 12 data bits as shown in Figure 47, Figure 48, and Figure 49. The AD5443 uses all 12 bits of DAC data. The AD5432 uses 10 bits and ignores the 2 LSBs, while the AD5426 uses 8 bits and ignores the last 4 bits.

Low Power Serial Interface

To minimize the power consumption of the device, the interface powers up fully only when the device is being written to, that is, on the falling edge of SYNC. The SCLK and DIN input buffers are powered down on the rising edge of SYNC. The SYNC of the AD5426/AD5432/AD5443 needs to be synchronous with the microprocessor control. Unfinished data frames are latched into the part and will affect the output.

DAC Control Bits C3 to C0

Control Bits C3 to C0 allow control of various functions of the DAC, as seen in Table 10. Default settings of the DAC on power-on are as follows: Data is clocked into the shift register on falling clock edges and daisy-chain mode is enabled. Device powers on with zero-scale load to the DAC register and I_{OUT} lines.

The DAC control bits allow the user to adjust certain features on power-on, for example, daisy-chaining may be disabled if not in use, active clock edge may be changed to rising edge, and DAC output may be cleared to either zero scale or midscale. The user may also initiate a readback of the DAC register contents for verification purposes.

Table 10. DAC Control Bits

C3	C2	C1	C0	Function Implemented
0	0	0	0	No operation (power-on default)
0	0	0	1	Load and update
0	0	1	0	Initiate readback
0	0	1	1	Reserved
0	1	0	0	Reserved
0	1	0	1	Reserved
0	1	1	0	Reserved
0	1	1	1	Reserved
1	0	0	0	Reserved
1	0	0	1	Daisy-chain disable
1	0	1	0	Clock data to shift register on rising edge
1	0	1	1	Clear DAC output to zero scale
1	1	0	0	Clear DAC output to midscale
1	1	0	1	Reserved
1	1	1	0	Reserved
1	1	1	1	Reserved

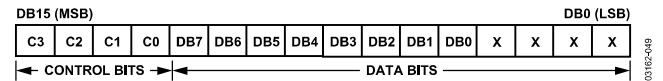


Figure 47. AD5426 8-Bit Input Shift Register Contents

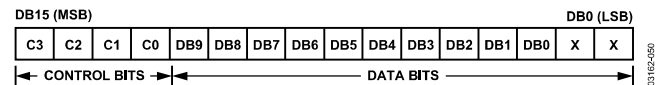


Figure 48. AD5432 10-Bit Input Shift Register Contents

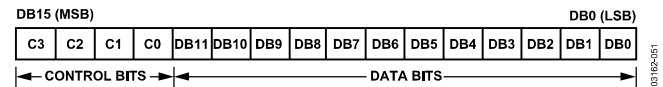


Figure 49. AD5443 12-Bit Input Shift Register Contents

SYNC Function

SYNC is an edge-triggered input that acts as a frame synchronization signal and chip enable. Data can be transferred into the device only while SYNC is low. To start the serial data transfer, SYNC should be taken low observing the minimum SYNC falling to SCLK falling edge setup time, t_{s1} .

Daisy-Chain Mode

Daisy-chain is the default power-on mode. Note that the SDO line operates with a V_{DD} of 3.0 V to 5.5 V. To disable the daisy chain function, write 1001 to the control word. In daisy-chain mode, the internal gating on SCLK is disabled. The SCLK is continuously applied to the input shift register when SYNC is low. If more than 16 clock pulses are applied, the data ripples out of the shift register and appears on the SDO line. This data is clocked out on the rising edge of SCLK (this is the default, use the control word to change the active edge) and is valid for the next device on the falling edge (default). By connecting this line to the D_{IN} input on the next device in the chain, a multidevice interface is constructed. Sixteen clock pulses are required for each device in the system. Therefore, the total number of clock cycles must equal 16 N where N is the total number of devices in the chain. See the timing diagram in Figure 4.

When the serial transfer to all devices is complete, SYNC should be taken high. This prevents any further data being clocked into the input shift register. A burst clock containing the exact number of clock cycles may be used and SYNC taken high some time later. After the rising edge of SYNC, data is automatically transferred from each device's input shift register to the addressed DAC.

When control bits = 0000, the device is in no operation mode. This may be useful in daisy-chain applications where the user does not want to change the settings of a particular DAC in the chain. Simply write 0000 to the control bits for that DAC and the following data bits will be ignored. To re-enable the daisy-chain mode, if disabled, a power recycle is required.